

FINAL EXAM (GRAPH MINING – SPRING 2026)

Full Name: _____

Student ID: _____

Instructions:

- Each response must include both the relevant formula(s) and a clear, **step-by-step** solution leading to the final answer.
- All answers must be written in **English**.

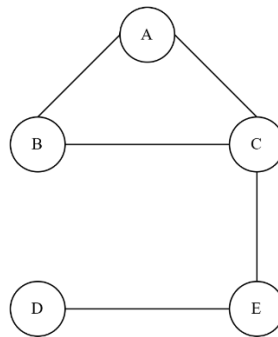
Question 1: (Graphlet Kernel) (10 points)

- a) Consider a small, undirected graph $G = (V, E)$ with 5 nodes where $V = A, B, C, D, E$, and $E = (A, B), (A, C), (B, C), (C, E), (E, D)$. Compute the graphlet count vector f_G for graphlets of size $k = 3$. (5 points)

1. g_1 : A path of length 2 (3 nodes, 2 edges)



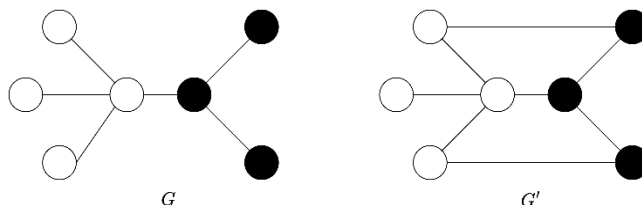
2. g_2 : A triangle (3 nodes, 3 edges)



- b) How does the choice of graphlet size k dictate the trade-off between the expressive power of the Graphlet Kernel and its computational complexity? (5 points)

Question 2: (Shortest Path Kernel) (5 points)

Consider the following two small undirected graphs with node labels (black and white)



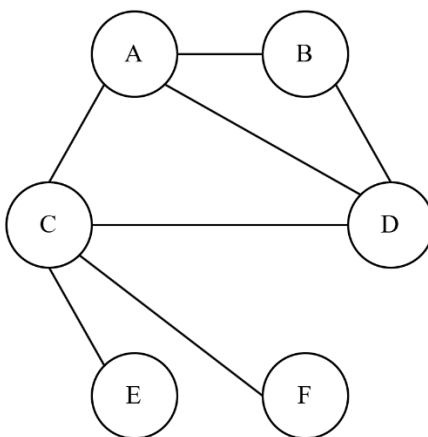
Given the shortest-path graph kernel is:

$$k_{\text{shortest paths}}(S_1, S_2) = \sum_{e_1 \in E_1} \sum_{e_2 \in E_2} k_{\text{walk}}^{(1)}(e_1, e_2)$$

- Compute all-pairs shortest path lengths for both graphs and compute the shortest path kernel value $K_{SP}(G, G')$. (3 points)
- What are the main theoretical and practical limitations of the shortest path kernel? (2 points)

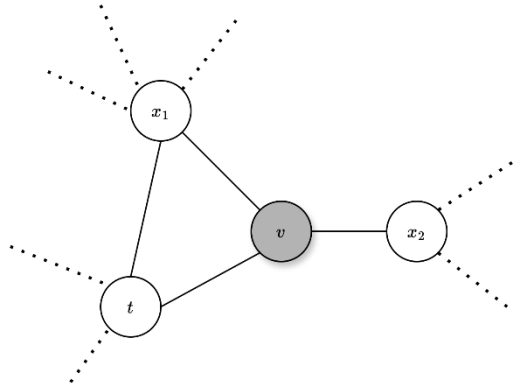
Question 3: (Weiseiler-Lehman Kernel) (10 points)

- Perform one iteration (color refinement) of the 1-WL test for the given graph. Initialize all nodes with the color “1”. Show the multi-set of neighborhood colors for each node, the aggregated string, and assign new hashed colors. (5 points)



- Provide an example of a simple graph structure that the 1-WL test is provably incapable of detecting or counting. (5 points)

Question 4: (Node2Vec) (10 points)



Consider an unweighted graph as provided, where a node2vec random walk has just traversed the edge from node t to node v . The walk is currently at node v and needs to decide its next step.

Suppose the node2vec parameters are set as follows:

- Return parameter: $p = 0.5$
- In-out parameter: $q = 2.0$

Given 2nd-order random walk transition probability formula:

$$\pi_{vx} = \alpha_{pq}(t, x) \cdot w_{vx}$$

where $w_{vx} = 1$ in case of unweighted graphs and

$$\alpha_{pq}(t, x) = \begin{cases} \frac{1}{p} & \text{if } d_{tx} = 0 \\ 1 & \text{if } d_{tx} = 1 \\ \frac{1}{q} & \text{if } d_{tx} = 2 \end{cases}$$

- Determine the unnormalized transition probabilities (π) for moving from node v to each of its three neighbors (t , x_1 , and x_2). (5 points)
- How do the parameters p and q allow Node2Vec to interpolate between Breadth-First Search (BFS) and Depth-First Search (DFS) behavior? What specific structural properties of the graph does each search strategy capture? (5 points)

Question 5: (Div2Vec) (5 points)

Given a small graph where node A is the current node in a random walk. Node A has three neighbors: X , Y , and Z . The degrees of these neighbor nodes in the overall graph are as follows:

- $\deg(X) = 4$
- $\deg(Y) = 16$
- $\deg(Z) = 16$

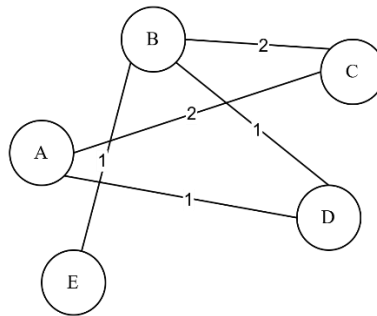
Known that, the probability of choosing the next node $v \in N(u)$ defined as:

$$P(v) = \frac{f(\deg(v))}{\sum_{x \in N(u)} f(\deg(x))}$$

and $f(x) = 1/x$

- a) Calculate the exact transition probabilities from node A to nodes X , Y , and Z . (3 points)
- b) What is the theoretical trade-off between using *standard div2vec* ($f(x) = 1/x$) versus *rooted div2vec* ($f(x) = 1/\sqrt{x}$) in the context of a recommender system? (2 points)

Question 6: (LINE) (10 points)



Given a small, undirected information network with 5 vertices, $V = \{A, B, C, D, E\}$. The connections (edges) between these vertices have specific weights representing the strength of their relationships.

Here are the observed edges and their weights (w):

- Edge (A, C) has a weight of 2
- Edge (A, D) has a weight of 1
- Edge (B, C) has a weight of 2
- Edge (B, D) has a weight of 1
- Edge (B, E) has a weight of 1

Assume all other unlisted vertex pairs have no direct edge, meaning a weight of 0. Also, assume the vertices do not connect to themselves, so $w_{A,A} = 0$.

- What is the **first-order** proximity between vertex A and vertex B? (3 points)
- Given that the second-order proximity relies on the empirical distribution of "contexts" for a given vertex, defined as $\hat{p}_2(v_j|v_i) = \frac{w_{ij}}{d_i}$, where d_i is the degree (sum of edge weights) of vertex i . Calculate the empirical probability $\hat{p}_2(C|A)$ and do vertex A and vertex B have a non-zero second-order proximity? (3 points)
- First-order proximity is only applicable to *undirected graphs*, whereas its formulation for second-order proximity can be applied to *both directed and undirected graphs*. What is the reason? (4 points)

Question 7: (HOPE) (10 points)

HOPE model approximates high-order proximities using a general formulation:

$$S = M_g^{-1} \cdot M_l$$

For the Adamic-Adar (AA) measurement, the proximity accounts for the degree of the intermediate nodes. A diagonal matrix D is used, where $D_{ii} = 1/\sum_j (A_{ij} + A_{ji})$. This means D_{ii} is the reciprocal of the total degree (in-degree + out-degree) of vertex v_i . So,

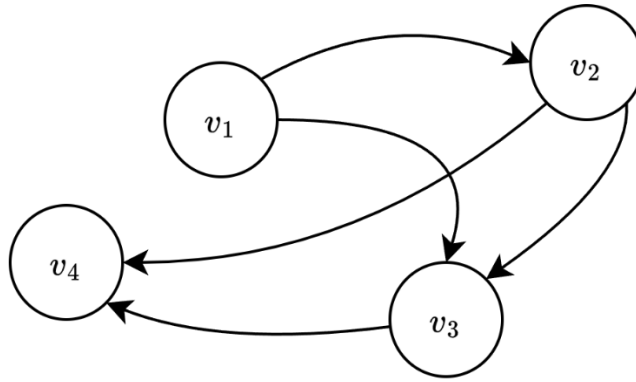
$$\mathbf{M}_g = \mathbf{I}$$

$$\mathbf{M}_l = \mathbf{A} \cdot \mathbf{D} \cdot \mathbf{A}$$

Suppose you are given a directed graph with 4 vertices (v_1, v_2, v_3, v_4) and the following adjacency matrix A :

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

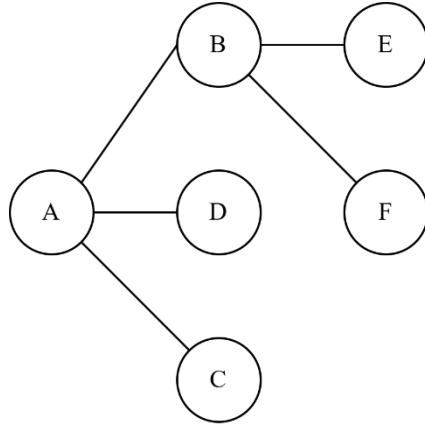
This graph has directed edges $v_1 \rightarrow v_2$, $v_1 \rightarrow v_3$, $v_2 \rightarrow v_3$, $v_2 \rightarrow v_4$, and $v_3 \rightarrow v_4$.



- Based Calculate the total degree for each vertex and construct the diagonal matrix D . (4 points)
- Calculate the exact high-order proximity matrix S for Adamic-Adar, given the adjacency matrix A and matrix D . (4 points)
- Why does the HOPE model avoid directly computing the standard SVD on the high-order proximity matrix S ? In your answer, explain how calculating S explicitly affects matrix density and the overall time complexity compared to the JDGSVD approach. (2 points)

Question 8: (struc2vec) (5 points)

Consider an undirected, unweighted graph G with a set of 6 nodes $V = \{A, B, C, D, E, F\}$



Given that

- $R_k(u)$ is the set of nodes at an exact distance of k hops from node u .
- $s(R_k(u))$ is the ordered degree sequence of that set.
- The structural distance at layer k , denoted as $f_k(u, v)$, is defined as:

$$f_k(u, v) = f_{k-1}(u, v) + g\left(s(R_k(u)), s(R_k(v))\right)$$

- Distance function:

$$d(a, b) = \frac{\max(a, b)}{\min(a, b)} - 1$$

- The weight of the edge between u and v in layer k of the context graph is:

$$w_k(u, v) = e^{-f_k(u, v)}$$

- Simplified DTW Calculation (g): When comparing a longer sequence (e.g., $[x, y, z]$) to a sequence of just one number (e.g., $[w]$), calculate the distance by aligning the single number to every number in the longer sequence. The total cost is $d(x, w) + d(y, w) + d(z, w)$.
 - a) What is the structural distance $f_0(A, C)$ and $f_1(A, C)$? (3 points)
 - b) Why do models like DeepWalk or node2vec struggle to identify two nodes as structurally equivalent if they are located far apart in a large network, and how does struc2vec bypass this limitation? (2 points)

Question 9: (Role2Vec) (5 points)

Consider a small graph $G = (V, E)$ consisting of four vertices: $V = \{v_1, v_2, v_3, v_4\}$. Instead of treating these vertices purely by their IDs, each vertex has an observed attribute/feature vector $x_i \in \mathbb{R}^2$:

$$x_1 = [1, 2]$$

$$x_2 = [3, 0]$$

$$x_3 = [0, 1]$$

$$x_4 = [2, 1]$$

To group these vertices into structural roles (vertex-types), we define a simple mapping function $\Phi(x)$ that uses the sum operator on the attribute vector:

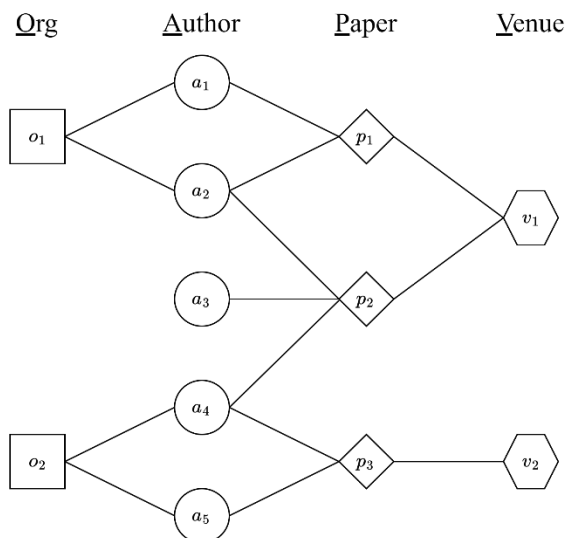
$$\Phi(x_i) = \sum_{k=1}^K x_{i,k}$$

Where $x_{i,k}$ is the k -th element of the attribute vector for vertex i

- Using the mapping function $\Phi(x)$, calculate the resulting vertex-type w for each of the four vertices. (2 points)
- Assume a traditional random walk algorithm traverses the graph and generates the following sequence of vertex IDs: $W = (v_3, v_1, v_4, v_2, v_1)$. Convert this standard random walk into a Role2Vec attributed random walk. (2 points)
- What are the advantages of using *node attributed embeddings* compared to traditional identity-based embeddings? (1 point)

Question 10: (metapath2vec) (5 points)

Consider the following heterogeneous academic network:



- List all valid APVPA meta-paths that start with author a_2 and end with a different author. (3 points)
- Meta-paths are typically designed to be symmetric (where the first node type is the same as the last node type, such as APVPA). What is the algorithmic necessity of this symmetry when generating paths for the skip-gram model? (2 points)

Question 11: (HIN2Vec) (10 points)

Given a small Heterogeneous Information Network (HIN) with a total of 4 nodes ($|V| = 4$). We want to embed these nodes into a 2-dimensional latent space ($d = 2$). The model is set up to predict 3 different meta-path relationships ($|R| = 3$).

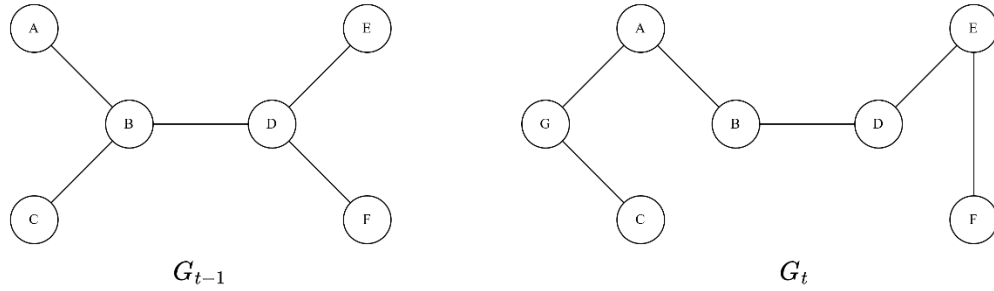
Suppose the trained transformation matrix for the source node, W_X , is given as:

$$W_X = \begin{bmatrix} 0.1 & 0.2 \\ 0.5 & 0.9 \\ 0.4 & 0.6 \\ 0.8 & 0.3 \end{bmatrix}$$

- What are the dimensions of the input vector $\vec{x}$, the transformation matrix W_X , and its transpose W_X' ? Write out the one-hot encoded vector $\vec{x}$. (3 points)
- According to the HIN2Vec framework, the latent vector for node x is computed as $W_X' \vec{x}$. Calculate this resulting latent vector step-by-step. (3 points)
- During the training data preparation phase of HIN2Vec, why is it crucial to explicitly filter out cycles (paths where the same node appears more than once) when generating random walks? (4 points)

Question 12: (dynnode2vec) (10 points)

Consider a dynamic network that evolves from timestamp $t - 1$ to timestamp t as follow:



- Identify the set of newly added nodes (V_{add}), set of newly added edges (E_{add}) and the set of deleted edges (E_{del}). (5 points)
- According to the dynnode2vec architecture, how exactly is the set of 'evolving nodes' (ΔV_t) defined, and under what conditions would the time efficiency of this generation step degrade to match that of static node2vec? (5 points)

Question 13: (Continuous-Time Dynamic Network Embeddings) (5 points)

Suppose you are applying the CTDNE framework to a small, continuous-time dynamic network $G = (V, E_T, \mathcal{T})$. The network contains exactly three temporal edges that occurred over a short timeframe:

Edge 1 (e_1): Node A interacts with Node B at timestamp $\mathcal{T}(e_1) = 1$

Edge 2 (e_2): Node B interacts with Node C at timestamp $\mathcal{T}(e_2) = 2$

Edge 3 (e_3): Node A interacts with Node C at timestamp $\mathcal{T}(e_3) = 3$

- Using the exponential biased distribution, calculate the probability of selecting e_2 as the initial edge. Given the Exponential Biased Selection Formula: (3 points)

$$\Pr(e) = \frac{\exp(\mathcal{T}(e) - t_{min}))}{\sum_{e' \in E_T} \exp(\mathcal{T}(e') - t_{min}))}$$

Note: You can leave your final answer as a fraction in terms of Euler's number (e)

- When choosing the very first edge to start a temporal random walk, how does applying the exponential bias distribution alter the behavior of the model compared to the unbiased (uniform) approach? (2 points)